

Part 4 of 5

Designing with FPGAs: Clock Management

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The focus in this article is on clock management in field-programmable gate arrays (FPGAs). It talks about the digital clock manager (DCM), which is a primitive used in FPGAs from Xilinx for dealing with all aspects of clock management. DCMs are used to implement delay-locked loop (DLL), digital frequency synthesiser, digital phase shifter (PS) and digital spread spectrum. They provide advanced clocking capabilities

ties to multiply or divide the incoming clock frequency to synthesise a new clock frequency. They also eliminate clock skew, thereby improving system performance, and are capable of phase shifting the output clock to delay the incoming clock by a fraction of clock period. Table 1 lists the advanced clocking capabilities of a DCM.

What is DCM

Fig. 1 shows the functional block diagram of a DCM. As shown in the figure, it comprises four different

functional units, namely, DLL, digital frequency synthesiser (DFS), PS and status logic (SL).

The DLL unit provides an on-chip digital deskew circuit that generates output clock signals having zero propagation delay. The input signals to the DLL unit are CLKIN and CLKFB. The output signals from the DLL are CLK0, CLK90, CLK180, CLK270, CLK2X, CLK2X180 and CLKDV. The deskew circuit compensates for the delay on the routing network by monitoring the output clock (either the CLK0 or the CLK2X).

The DFS unit provides a wide and flexible range of output frequencies based on the ratio of two user-defined integers, a multiplier (CLKFX_MULTIPLY) and a divisor (CLKFX_DIVIDE). The output frequency is derived from the input clock (CLKIN) by simultaneous frequency division and multiplication.

The PS unit controls the phase relations of the DCM's clock outputs with respect to the CLKIN input. It shifts the phase of all nine DCM clock output signals by a fixed fraction of the input clock period. The fixed phase-shift value is set at design time and loaded

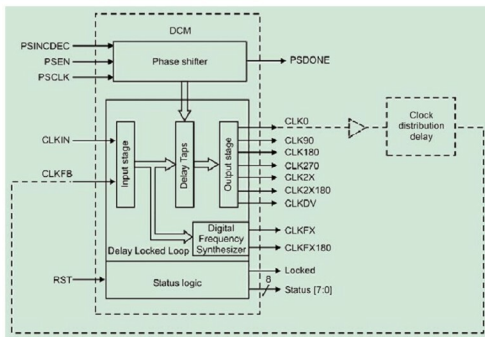


Fig. 1: Functional block diagram of DCM

TABLE 1
Capabilities of a Digital Clock Manager

Functionality	Application
Multiply or divide an incoming clock frequency	Synthesise a new frequency by clock multiplication and division
Clock conditioning	Ensuring a clean clock with 50 per cent duty cycle
Variable or fixed phase shifting of clock	Clock delay
Eliminate clock skew	To eliminate clock distribution delays and to improve overall system performance
Mirror, forward or buffer a clock signal	Deskew the clock and convert it to a different input/output standard

into the DCM during FPGA configuration.

The PS unit also provides a digital interface for the FPGA application to dynamically advance or retard the

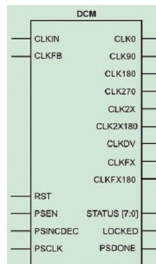


Fig. 2: DCM primitive